

KV Responsible AI

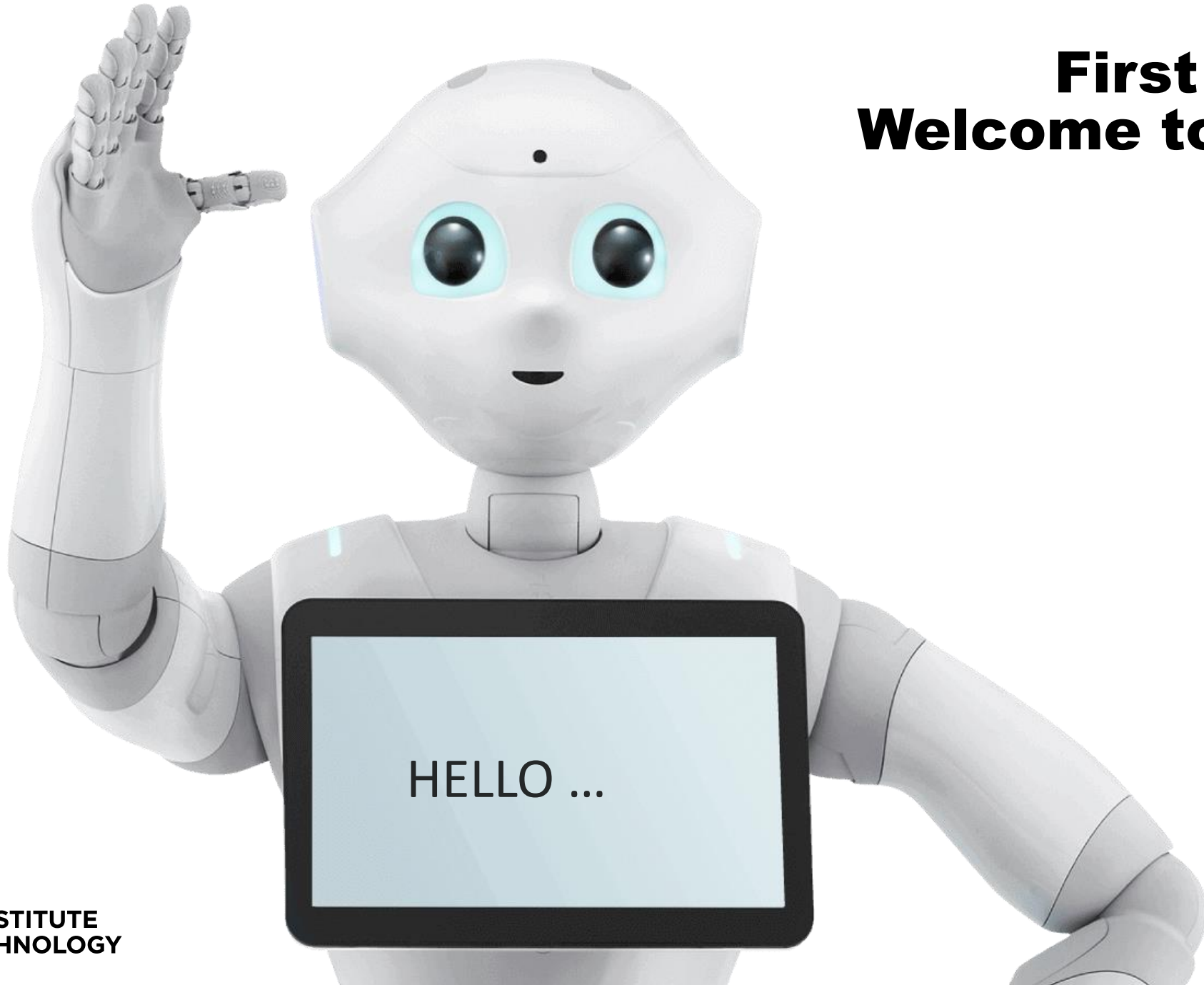


Martina Mara & Sandra Siedl

**Introduction
09.10.2023**



**First of all:
Welcome to JKU!**



Short intro Martina Mara



Austrian Council on Robotics and Artificial Intelligence

Vienna, November 2018

Shaping the Future of Austria
with Robotics and Artificial
Intelligence

White Paper by the Austrian Council
on Robotics and Artificial Intelligence



Prof. Dr. Martina Mara,
Head of LIT Robopsychology Lab



Short intro Sandra Siedl



Sandra Maria Siedl, BA MA,
LIT Robopsychology Lab

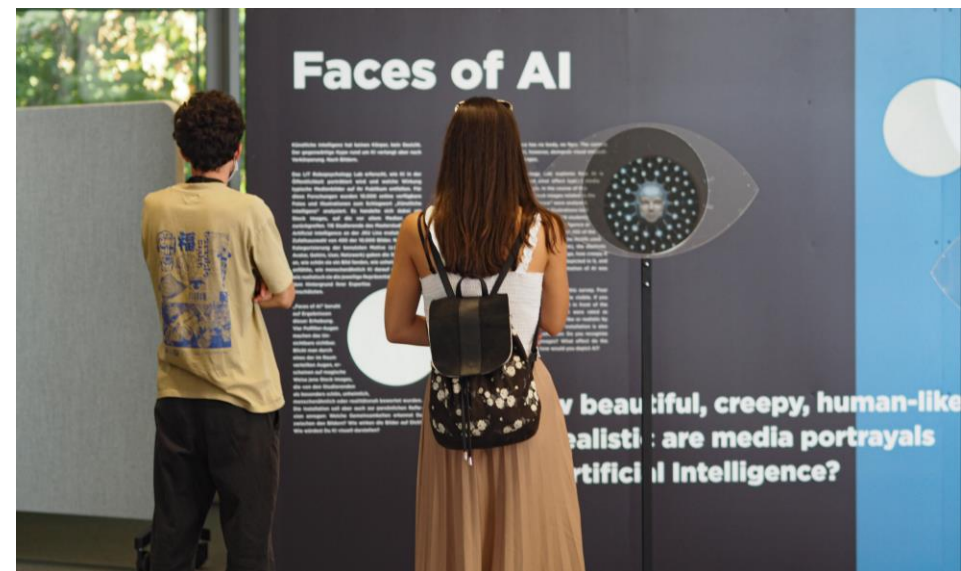
What is Robopsychology?

- Putting human experience and needs at the center of technological development
- Applying scientific methods and theories from Psychology to human interaction with Artificial Intelligence and Robotics
- Practical direction: Creating evidence-based guidelines for human-centered technology, collaborating with other disciplines & industry

Exemplary research questions:

- How is system understandability related to trust?
- How can we people collaborate well with algorithmic decision support systems?
- What does it take for robots not to undermine psychological needs of users, e.g. autonomy?
- How are traditional gender roles reflected in our interactions with AI assistants or robots?
- When do we see chatbots as humanlike and what are consequences thereof?

Science communication: Demystify AI @ Ars Electronica 2021



Where you find us on campus



Course overview: Online tools we are going to use

- **KUSSS** (kusss.jku.at) for
 - Course registration
 - Exam registration
 - Final grades
 - Sometimes emails to students
- **MOODLE** (moodle.jku.at) for
 - Q&A Forum
 - Session live streams
 - Video recordings of sessions
 - Slides and other course materials
 - Group assignment for group task
 - Submission of mini tasks (individual) & group task (in groups)
 - Final exam

Course overview: Online tools we are going to use

- **MENTIMETER** (menti.com) for
 - Live polls during sessions

Let's try Mentimeter

What motivated you to study AI?

Scan QR code or go to **menti.com** and type in:

2216 1789



What does responsibility mean?

Speaking of "**Responsible AI**" ... what does "**responsible**" actually mean?

to be responsible for sth

to have control and authority over something and the duty of taking care of it

responsible *adjective*

having good judgment and the ability to act correctly and make decisions on your own

- *a hard-working and responsible employee*
- *Let's stay calm and try to behave like responsible adults.*
- *Many big companies are now becoming more responsible.*

Source: Cambridge Dictionary

Parties that can be responsible in the AI sphere

Responsibilities in the development and implementation of AI can vary between parties:

Developers/engineers: Responsible for designing and training AI models that are technically sound, ethical, and that minimize undesirable consequences.

Data collectors/providers: Must ensure that the data is high quality (e.g. representative, ethical/fair).

Regulators: Responsible for creating and enforcing laws governing the development and use of AI to protect society's interests.

Companies/providers: Must ensure that AI systems are used in accordance with their business objectives, with regulations and ethical standards.

Users/society: Have a responsibility to understand the implications of AI systems, advocate for their ethical use, and report misuses.

Researchers from other disciplines/ethics experts: Help define guidelines and frameworks for responsible development and use of AI systems.

AI systems themselves (?): Even if AI has no accountability itself, it is important that AI systems are designed to follow laws and ethical principles, and that they are responsive to human oversight.

Companies: RBC & CSR

According to the OECD's understanding, **Responsible business conduct (RBC)** means that businesses

- a) should make a positive contribution to economic, environmental and social progress with a view to achieving sustainable development and
- b) should avoid and address adverse impacts through their own activities and prevent or mitigate adverse impacts directly linked to their operations, products or services by a business relationship. Risk-based due diligence is central to identifying, preventing and mitigating actual and potential adverse impacts, and thus is a key element of RBC.

(OECD Policy Framework for Investment, 2015)

According to the European Commission's understanding, **Corporate Social Responsibility (CSR)** is

the process whereby enterprises integrate social, environmental, ethical and human rights concerns into their core strategy, operations and integrated performance, in close collaboration with their stakeholders.

(EC Communication, 2011)

Individuals: Socially Responsible Personality

Psychologists began working on questionnaires to assess socially responsible personalities many years ago. Berkowitz & Lutterman (1968) proposed the following scale:

Answer each item with:

Strongly agree (5), Agree (4), Undecided (3), Disagree (2), or Strongly disagree (1)

1. It is no use worrying about current events or public affairs; I can't do anything about them anyway.
2. Every person should give some of their time for the good of their town or country.
3. Our country would be a lot better off if we didn't have so many elections and people didn't have to vote so often.
4. Letting your friends down is not so bad because you can't do good all the time for everybody.
5. It is the duty of each person to do their job the very best they can.
6. People would be a lot better off if they could live far away from other people and never have to do anything for them.
7. At school I usually volunteered for special projects.
8. I feel very bad when I have failed to finish a job I promised I would do.

What does AI mean?

Speaking of "**Responsible AI**" ... what does "**AI**" actually mean?

- Artificial intelligence (AI) refers to systems that display “intelligent” behavior by analyzing their environment and taking actions – with some degree of autonomy – to achieve specific goals.
- Many current AI technologies (probabilistic ones such as ML, NN, DL) use large datasets, recognize patterns in them and automate decision-making in a certain domain.

Some application examples:

- AI-based systems may include voice assistants, image recognition software, search engines, translation services or face recognition systems. We use AI on a daily basis, e.g. to translate languages, generate subtitles in videos or to block email spam.
- AI can also be embedded in hardware like autonomous cars, drones, or advanced industrial robots.

AI is human-made!



Artificial Intelligence is designed by us:

- Which goals an AI system should target is defined by people.
- Where and how an AI system will be applied is defined by people.
- On which data an AI system is trained is defined by people.
- The data, from which we let AI learn, are often created by humans (a.o., past decisions, texts, images, videos)

Accordingly, **we humans are also responsible for shaping AI** in a way that

- respects fundamental human rights and ethical norms,
- maintains human autonomy and control,
- no one is being discriminated against,
- as many people as possible can benefit from AI.

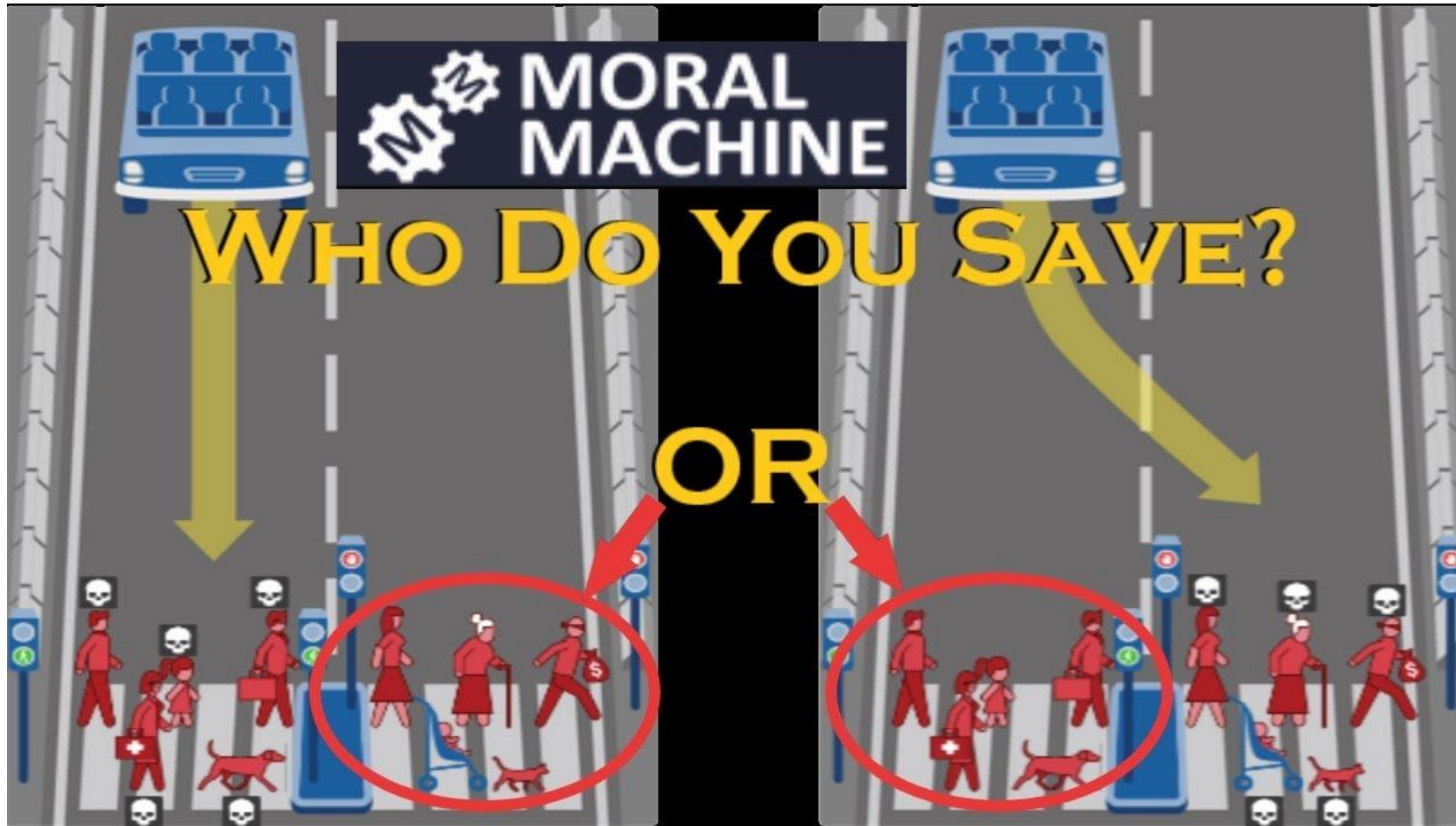
Main goals of this course

- Discussion of current ethical, social, and environmental questions of Artificial Intelligence
- Learn about state-of-the-art international guidelines for trustworthy/responsible AI
- Think about what you can consider as a (future) developer/engineer to create responsible AI
- Learn about ethical decision-making and taking different perspectives on AI
- Discussion of positive future scenarios with AI (“AI for the good”, “AI for all”)

Planned topics of this course

- International Guidelines for Responsible/Ethical/Trustworthy AI
- AI and environmental wellbeing
- Moral decision-making, different schools of ethics
- Biased AI & fairness
- Understandable and explainable AI
- Media depictions of AI
- AI and misinformation
- AI and privacy
- Social AI/robots & social engineering
- Diversity and interdisciplinarity in teams that create AI

Quick teaser: Ethics in AI decision-making



Quick teaser: (Un)fair AI?



The New York Times

There Is a Racial Divide in Speech-Recognition Systems, Researchers Say

Technology from Amazon, Apple, Google, IBM and Microsoft misidentified 35 percent of words from people who were black. White people fared much better.

future tense

When Alexa Can't Understand You

People with speech disabilities are being left out of the voice-assistant revolution.

By MOIRA CORCORAN

OCT 16, 2018 • 7:30 AM

Course schedule

↑ Datum	Uhrzeit	Raum	Thema
Mo. 09.10.2023	12:00 - 13:30	HS 7	Introduction
Mo. 16.10.2023	12:00 - 13:30	HS 7	
Mo. 23.10.2023	12:00 - 13:30	HS 7	
Mo. 30.10.2023	12:00 - 13:30	HS 7	
Mo. 06.11.2023	12:00 - 13:30	HS 7	
Mo. 13.11.2023	12:00 - 13:30	HS 7	
Mo. 20.11.2023	12:00 - 13:30	HS 7	
Mo. 27.11.2023	12:00 - 13:30	HS 7	
Mo. 04.12.2023	12:00 - 13:30	HS 7	
Mo. 11.12.2023	12:00 - 13:30	HS 7	
Mo. 08.01.2024	12:00 - 13:30	HS 7	
Mo. 15.01.2024	12:00 - 13:30	HS 7	
Mo. 22.01.2024	12:00 - 13:30	HS 7	
Mo. 29.01.2024	12:00 - 13:30	HS 7	Written exam

The course takes place weekly physically in HS7 and is streamed live to JKU's lecture halls in Vienna and Bregenz.

There is also a Zoom live stream (link in Moodle).

Live attendance (in person or online) is required on at least 50% of the sessions from October 16 to January 22.

The **written test** will be implemented as a **Moodle test** and will take place under supervision **in the lecture hall**.

Course requirements

- **Mandatory elements**

- **50% live attendance** (in person or online) = at least 6 out of 12 upcoming sessions
- **Written exam** (on **29.01.2024**, multiple choice questions + 2 open questions)

- **Optional elements**

- **Group exercise** (“ethical dilemma discussion” in groups of 8 people)
- **3 mini tasks** (individual submission)

Course grading

Scoring scheme

Total points that can be achieved in the course: **40**

Group exercise: max. **12** points

Written exam: max. **22** points (**at least 11 points must be achieved**, otherwise repeat/fail)

3 mini tasks: **2** points each (if completed within 2 weeks after announcement)

40 - 35 points =	sehr gut (1)
< 35 points =	gut (2)
< 30 points =	befriedigend (3)
< 25 points =	genügend (4)
< 20 points =	nicht genügend (5)

If you hit the 40 points,
you'll get a fancy
unicorn sticker!



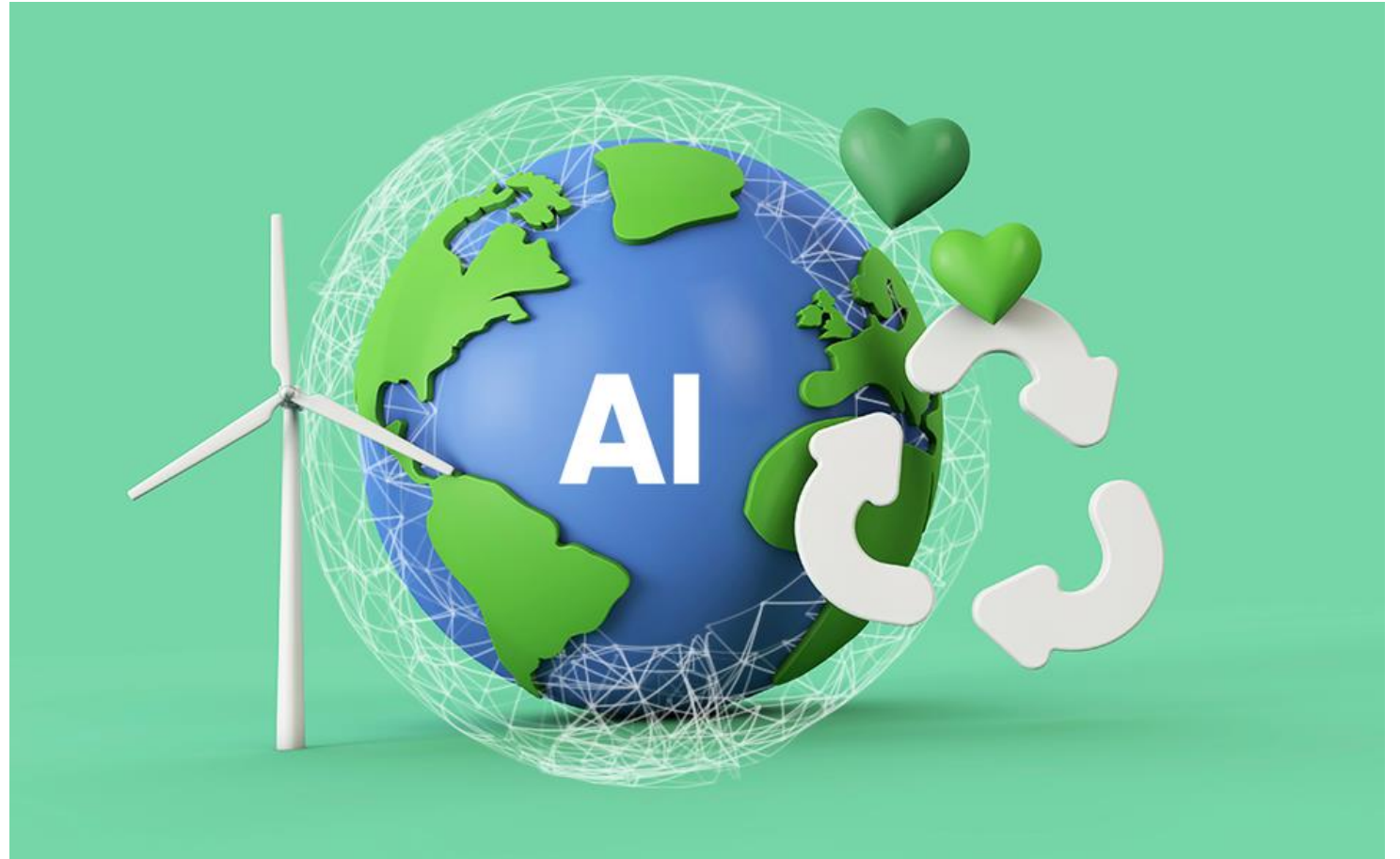
Furthermore important:

- Check course content regularly and do not miss deadlines (e.g., self-assignment to group, task submissions)!
- Feel free to engage in conversations about the course topics, but always remain respectful!

Outlook:

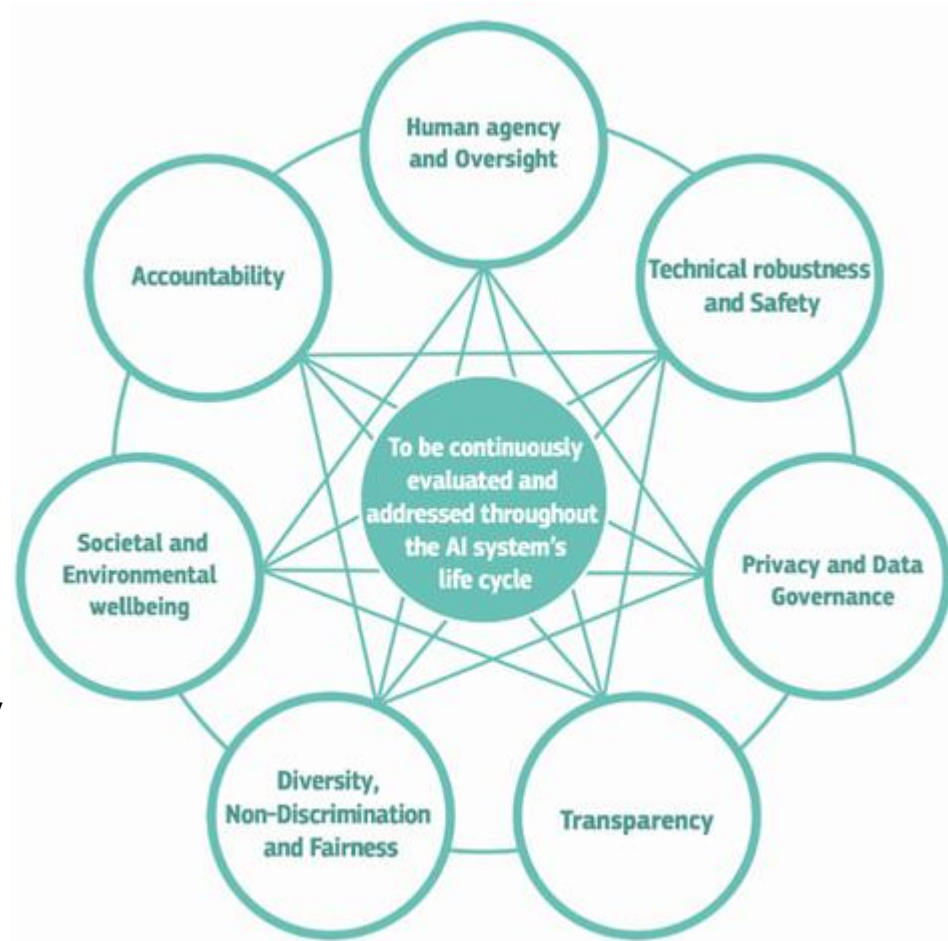
Next week:

- AI & Climate Change
- Instruction of mini task 1



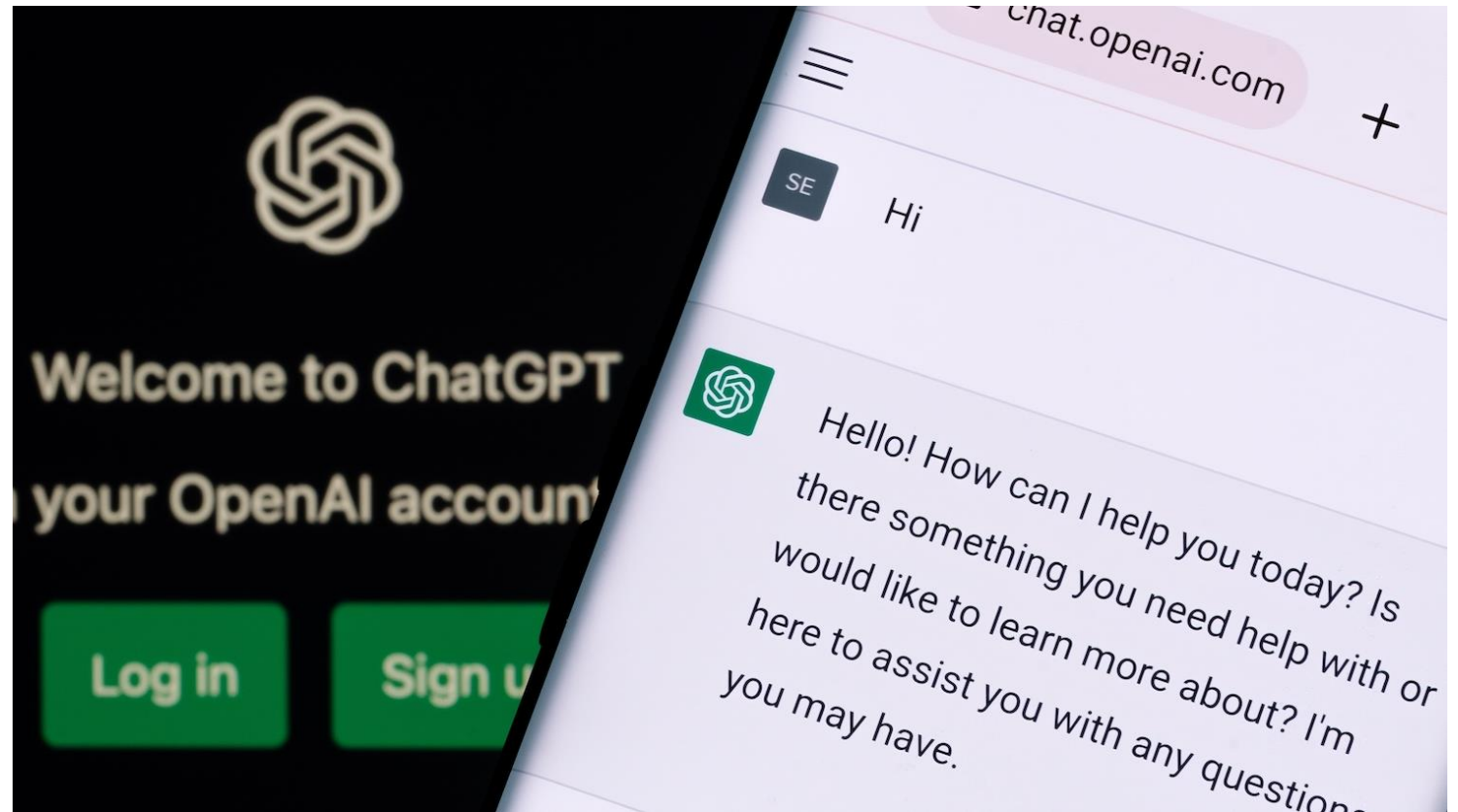
EU Ethics Guidelines for Trustworthy AI

“Societal and environmental wellbeing” is named one of seven key factors of “Trustworthy AI” in the European Union’s Ethics Guidelines.



Question to think about until next week:

If you had to calculate the amount of energy it took to answer your last ChatGPT prompt, what factors would you include in that calculation?



Questions?

KV Responsible AI



See you next week!

